

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO2 ASSESSMENT TOOL 1 — SCENARIO-BASED REQUIREMENT ANALYSIS

Software Requirement Specification (SRS)

Intelligent AI-Based Smart Public Transportation Demand Forecasting System

Field	Details
Course Code	CSA10
Course Title	Software Engineering
CO Assessed	CO2
Assessment Tool	Assessment Tool 1 — Scenario-Based Requirement Analysis
Weightage	-
Student Name	N.RAMYA
Register Number	192511403
Date	4/08/2026
Duration	-



Table of Contents

1. Introduction
2. Problem Description
3. Scope of the System
4. Functional Requirements
5. Non-Functional Requirements
6. Actors and Use Cases
7. Data Requirements
8. External Interface Requirements
9. System Features
10. Assumptions and Constraints
11. Requirement Validation — Completeness and Consistency

Problem Statement

Urban public transportation systems often struggle to efficiently manage passenger demand due to changing travel patterns, weather conditions, traffic congestion, and special events. Existing scheduling methods primarily rely on fixed timetables and historical assumptions, resulting in overcrowded vehicles during peak hours, underutilized fleets during off-peak periods, increased passenger waiting times, and inefficient resource allocation.

To address these challenges, an AI-powered Transportation Demand Forecasting Platform is proposed. The system will analyze historical passenger data, weather information, traffic conditions, public events, and real-time transportation data to accurately predict future passenger demand. Based on these predictions, the platform will recommend optimized bus and train schedules, improve fleet allocation, provide real-time passenger updates, and generate analytical insights for transport authorities. This intelligent approach will enhance operational efficiency, reduce congestion, improve commuter experience, and support sustainable urban transportation management.

1. INTRODUCTION

1.1 Purpose

This Software Requirement Specification (SRS) document defines the functional and non-functional requirements for the Intelligent AI-Based Smart Public Transportation Demand Forecasting System. The purpose of the system is to help public transportation agencies predict passenger demand accurately, optimize schedules and fleet allocation, and provide real-time passenger information using Artificial Intelligence (AI), the Internet of Things (IoT), Global Positioning System (GPS) tracking, and Big Data Analytics. This document is intended for use by the development team, transit agency stakeholders, quality assurance engineers, and project evaluators as the single source of truth for what the system must do.

1.2 Business Domain

The system operates in the Public Transportation / Smart Mobility domain, specifically addressing urban and metropolitan bus and rail transit operations. This domain is characterized by time-sensitive service delivery, high variability in commuter demand, dependency on external conditions such as weather and traffic, and a strong public interest in service reliability, safety, and affordability.

1.3 Stakeholders

Stakeholder	Role / Interest in the System
Passengers / Commuters	Primary end users who rely on accurate arrival times, reduced overcrowding, and reliable service.
Transit Operators / Schedulers	Responsible for planning and adjusting bus/train schedules based on AI-generated demand forecasts.
Fleet Managers	Manage vehicle allocation, maintenance windows, and driver assignment based on optimized fleet plans.
Transportation Agency Management	Oversee service quality, budget, and long-term policy decisions supported by analytics.
System Administrators	Maintain system configuration, user accounts, security, and system health.
Third-Party Data Providers	Supply external weather, traffic, and event data consumed by the forecasting engine.
Municipal / Regulatory Authorities	Set compliance, safety, and reporting requirements the system must support.

1.4 Existing Challenges

- Overcrowding of buses and trains during peak commuting hours.
- Underutilized vehicles and wasted operating costs during off-peak periods.
- Static, manually created schedules that cannot adapt to real-time conditions.
- Lack of visibility into how weather, local events, and traffic conditions affect ridership.
- Limited real-time information available to passengers, leading to poor commuter experience.
- Inefficient fleet and driver allocation due to the absence of predictive planning tools.

1.5 System Objectives

1. Forecast passenger demand using AI models trained on historical and real-time data.
2. Optimize public transport schedules dynamically based on predicted demand.
3. Improve fleet allocation and utilization across routes and time periods.
4. Monitor passenger flow in real time using IoT and GPS data.
5. Analyze travel patterns to support long-term route planning.
6. Generate transportation analytics and reports for decision-makers.
7. Reduce overcrowding during peak hours and improve off-peak utilization.
8. Improve overall public transport efficiency, reliability, and commuter satisfaction.

2. PROBLEM DESCRIPTION

Public transportation systems frequently experience overcrowding during peak hours and underutilization during off-peak periods because current scheduling approaches rely on static timetables and historical averages rather than predictive, real-time analysis. Traditional systems lack the ability to anticipate demand spikes caused by weather changes, public events, holidays, or unexpected traffic disruptions. As a result, transit agencies struggle to allocate the right number of vehicles to the right routes at the right times, leading to passenger dissatisfaction, inefficient resource usage, and increased operational costs.

The proposed Intelligent AI-Based Smart Public Transportation Demand Forecasting System addresses this gap by ingesting historical ridership data, live weather conditions, scheduled events, and real-time traffic patterns into an AI forecasting engine. The system will translate these predictions into actionable schedule and fleet recommendations, while also giving passengers real-time visibility into expected vehicle arrival times and crowding levels.

3. SCOPE OF THE SYSTEM

3.1 In Scope

- Collection and integration of historical ridership, GPS/IoT, weather, traffic, and event data.
- AI/ML-based demand forecasting at route, stop, and time-slot granularity.
- Schedule optimization recommendations for bus and train services.
- Fleet allocation optimization across routes and depots.
- Real-time passenger flow monitoring via IoT sensors and GPS-equipped vehicles.
- Passenger-facing mobile/web application for real-time arrival and crowding information.
- Dashboards for transit operators, fleet managers, and administrators.
- Analytics and reporting module for transportation performance metrics.

3.2 Out of Scope

- Ticketing, fare collection, and payment gateway processing.
 - Vehicle manufacturing, hardware procurement, or physical IoT device installation.
 - Autonomous vehicle control or driver-replacement automation.
 - Traffic signal control or direct integration with municipal traffic light systems.
 - Road infrastructure planning and construction management.
-

4. Functional Requirements

The following functional requirements define the specific behaviors and capabilities the system must provide, derived directly from the stated objectives of forecasting demand, optimizing schedules, improving fleet allocation, monitoring passenger flow, analyzing travel patterns, and generating analytics.

ID	Requirement	Description
FR-01	Demand Forecasting	The system shall analyze historical ridership, weather, event, and traffic data to forecast passenger demand per route, stop, and time interval.
FR-02	Schedule Recommendation	The system shall generate optimized bus/train schedule recommendations based on forecasted demand.
FR-03	Fleet Allocation	The system shall recommend optimal fleet (vehicle) allocation across routes and depots based on predicted demand and vehicle availability.
FR-04	Real-Time Passenger Flow Monitoring	The system shall collect and display real-time passenger flow and vehicle occupancy data using IoT sensors and GPS feeds.
FR-05	Real-Time Passenger Information	The system shall provide passengers with real-time arrival predictions and expected crowding levels via a mobile/web application.
FR-06	Travel Pattern Analysis	The system shall analyze historical and real-time travel data to identify recurring ridership patterns and trends.
FR-07	Analytics & Reporting	The system shall generate configurable transportation analytics reports (e.g., ridership trends, on-time performance, fleet utilization) for authorized users.
FR-08	External Data Integration	The system shall integrate with third-party weather and traffic APIs to enrich the forecasting model.
FR-09	Alert & Notification Management	The system shall send real-time alerts to passengers and operators regarding delays, overcrowding, or schedule changes.
FR-10	User & Role Management	The system shall allow administrators to create, modify, and deactivate user accounts with role-based access control.
FR-11	Historical Data Management	The system shall allow authorized users to import, store, and archive historical ridership and operations data.
FR-12	Feedback Collection	The system shall allow passengers to submit feedback on service quality, which shall be logged for analysis.
FR-13	Model Retraining	The system shall support periodic retraining/updating of the AI forecasting model using newly collected data.
FR-14	Audit Logging	The system shall log key user and system actions (schedule changes, fleet reallocations, admin actions) for auditing purposes.

5. NON-FUNCTIONAL REQUIREMENTS

Non-functional requirements define the quality attributes and operational constraints that the system must satisfy in addition to its functional capabilities.

Category	Requirement
Performance	The system shall generate demand forecasts for a given route within 5 seconds and refresh real-time dashboards at least every 30 seconds.
Scalability	The system shall support scaling to at least 10,000 concurrent passenger app users and data ingestion from 5,000+ IoT/GPS-enabled vehicles without service degradation.
Security	The system shall enforce role-based access control, encrypt data in transit (TLS 1.2+) and at rest (AES-256), and comply with applicable data protection regulations.
Reliability	Core forecasting and scheduling services shall maintain 99.5% uptime, with automatic failover for critical components.
Availability	The passenger-facing application and real-time information services shall be available 24/7, including during maintenance windows via redundant deployment.
Usability	The passenger and operator interfaces shall be designed for ease of use, achieving a System Usability Scale (SUS) score of 75 or higher in usability testing.
Maintainability	The system shall be built using modular, loosely coupled services to allow independent updates to the forecasting model, scheduling engine, and UI layers.
Interoperability	The system shall expose well-documented REST APIs to integrate with external weather, traffic, and municipal transit data sources.
Accuracy	The AI forecasting model shall achieve a minimum demand prediction accuracy (e.g., $MAPE \leq 15\%$) validated against historical ridership data.

Data Retention & Auditability	The system shall retain historical ridership and operational data for a minimum of 3 years and maintain immutable audit logs of administrative action.
Portability	The passenger application shall be accessible via both Android/iOS mobile apps and responsive web browsers.
Disaster Recovery	The system shall support automated daily backups with a recovery point objective (RPO) of 24 hours and a recovery time objective (RTO) of 4 hours.

6. ACTORS AND USE CASES

6.1 Identified Actors

Actor	Type	Description
Passenger / Commuter	Primary	View real-time arrival information; view predicted crowding level; submit service feedback.
Transit Operator / Scheduler	Primary	Generate optimized schedules; monitor passenger flow; analyze travel patterns; generate analytics reports.
Fleet Manager	Primary	Allocate fleet resources; manage historical ridership data used for fleet planning.
System Administrator	Primary	Manage users and system configuration; monitor system performance.
Weather Data Service (External)	Secondary / External System	Supplies real-time and forecast weather data feeds.
Traffic Data Service (External)	Secondary / External System	Supplies real-time traffic condition feeds.
GPS / IoT Sensor Network (External)	Secondary / External System	Supplies live vehicle location and passenger occupancy data.
AI Forecasting Engine (External/Internal Component)	Secondary / System Actor	Executes demand forecasting models and returns predictions; triggers real-time alerts.

6.2 Use Cases per Actor

Passenger / Commuter

- View Real-Time Arrival Information
- View Predicted Crowding Level
- Submit Service Feedback

Transit Operator / Scheduler

- Generate Optimized Schedule (includes Forecast Passenger Demand)
- Monitor Passenger Flow in Real Time

- Analyze Travel Patterns
- Generate Transportation Analytics Reports

Fleet Manager

- Allocate Fleet Resources (includes Generate Optimized Schedule)
- Manage Historical Ridership Data

System Administrator

- Manage Users and System Configuration
- Monitor System Performance

External / System Actors

- Ingest Weather Data Feed (Weather Data Service)
- Ingest Traffic Data Feed (Traffic Data Service)
- Ingest GPS / IoT Vehicle Data (GPS / IoT Sensor Network)
- Forecast Passenger Demand / Send Real-Time Passenger Alerts (AI Forecasting Engine)

6.3 Use Case Diagram

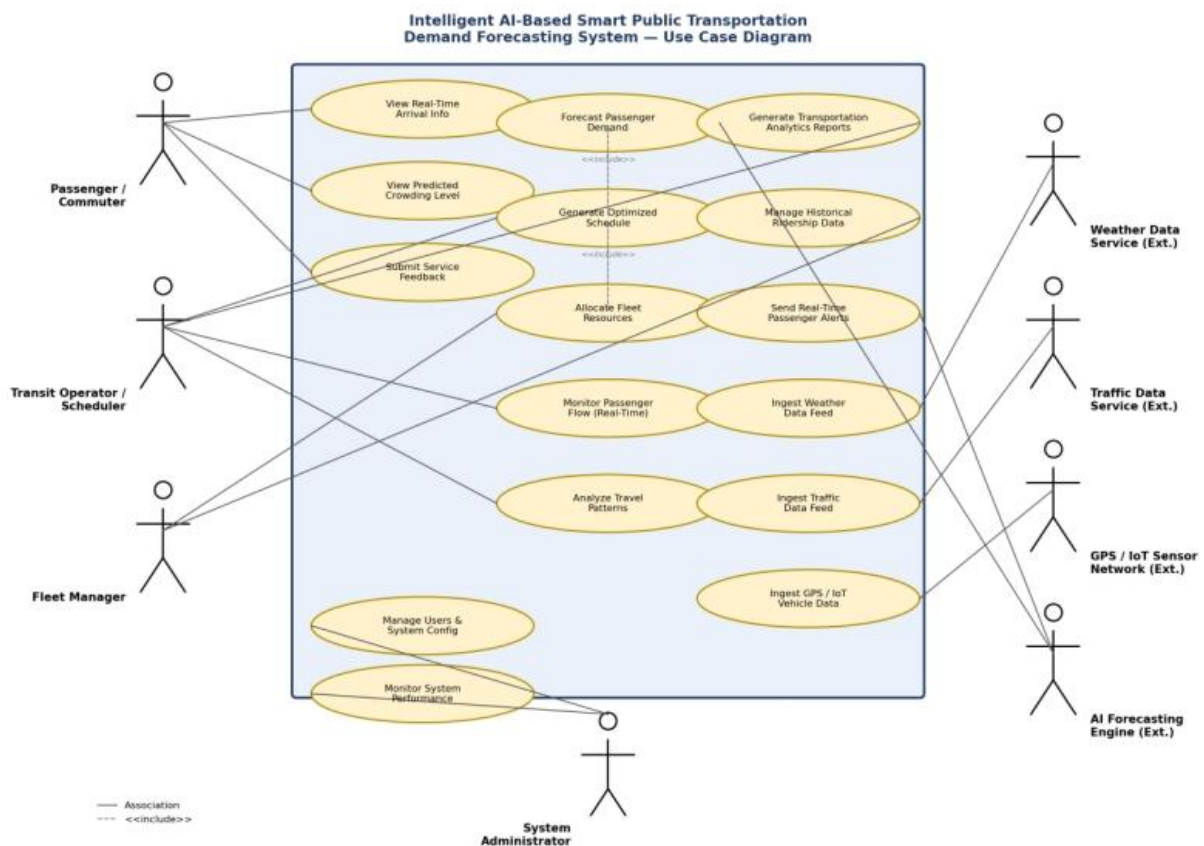


Figure 1: Use Case Diagram — Intelligent AI-Based Smart Public Transportation Demand Forecasting System

6.4 Detailed Use Case Descriptions

UC-01: Forecast Passenger Demand

Field	Details
Use Case ID	UC-01
Use Case Name	Forecast Passenger Demand
Actors	Transit Operator/Scheduler (primary), AI Forecasting Engine (system actor)
Preconditions	Historical ridership data and current weather/traffic/event feeds are available in the system.
Main Flow	1. Operator requests a demand forecast for a route and time window. 2. System retrieves historical ridership, weather, traffic, and event data. 3. AI Forecasting Engine processes the data and generates a demand prediction. 4. System displays the forecast with a confidence indicator.
Alternate / Exception Flow	If required external data (e.g., weather feed) is unavailable, the system shall fall back to historical-pattern-only forecasting and flag the result as lower confidence.
Postconditions	Forecast is generated, stored, and made available to the Schedule Optimization use case.

UC-02: Generate Optimized Schedule

Field	Details
Use Case ID	UC-02
Use Case Name	Generate Optimized Schedule
Actors	Transit Operator/Scheduler (primary); includes UC-01

Preconditions	A valid demand forecast exists for the selected route(s) and time period.
Main Flow	1. Operator selects route(s) and planning horizon. 2. System invokes Forecast Passenger Demand (UC-01). 3. Scheduling engine computes recommended trip frequency and timing. 4. Operator reviews and approves or edits the recommended schedule. 5. System publishes the approved schedule.
Alternate / Exception Flow	If the operator rejects the recommendation, the system shall allow manual schedule adjustment and log the override reason.
Postconditions	An approved, optimized schedule is published and made available to Fleet Manager and passenger app.

UC-03: Allocate Fleet Resources

Field	Details
Use Case ID	UC-03
Use Case Name	Allocate Fleet Resources
Actors	Fleet Manager (primary); includes UC-02
Preconditions	An approved optimized schedule exists; vehicle and driver availability data is current.
Main Flow	1. Fleet Manager opens the fleet allocation view. 2. System proposes vehicle-to-route assignments based on the optimized schedule and vehicle availability. 3. Fleet Manager reviews, adjusts, and confirms the allocation. 4. System updates fleet assignment records and notifies drivers/depots.
Alternate / Exception Flow	If insufficient vehicles are available, the system shall flag the shortfall and suggest the least-impact route reductions.
Postconditions	Fleet allocation is confirmed and reflected in operational dispatch records.

UC-04: View Real-Time Arrival Information

Field	Details
Use Case ID	UC-04
Use Case Name	View Real-Time Arrival Information
Actors	Passenger/Commuter (primary)
Preconditions	Passenger has the mobile/web app open and selected a route or stop.
Main Flow	1. Passenger selects a stop or route. 2. System retrieves live GPS vehicle position and predicted arrival time. 3. System displays estimated arrival time and current crowding level.
Alternate / Exception Flow	If GPS data for a vehicle is temporarily unavailable, the system shall display the last known estimate with a 'delayed data' indicator.
Postconditions	Passenger receives up-to-date arrival and crowding information.

7. DATA REQUIREMENTS

The system depends on the timely and accurate collection, storage, and processing of the following categories of data. All personally identifiable passenger data shall be handled in accordance with applicable data privacy regulations.

Data Category	Key Data Elements	Source
Historical Ridership Data	Route ID, stop ID, timestamp, boarding/alighting counts, trip ID	Internal ticketing/AFC systems, manual import
Real-Time GPS/IoT Data	Vehicle ID, latitude/longitude, speed, occupancy sensor count, timestamp	On-board GPS and IoT occupancy sensors
Weather Data	Location, temperature, precipitation, visibility, forecast timestamp	Third-party Weather Data API
Traffic Data	Route/segment ID, congestion level, average speed, incident flags	Third-party Traffic Data API
Event / Calendar Data	Event name, location, expected attendance, date/time	Municipal event calendars, manual entry
Forecast Output Data	Route ID, time slot, predicted demand, confidence score	Generated internally by AI Forecasting Engine
Schedule & Fleet Data	Trip ID, vehicle ID, driver ID, route ID, scheduled/actual times	Scheduling and Fleet Management modules
User Account Data	User ID, role, name, contact information, access permissions	Admin & User Management Service
Feedback Data	Passenger ID (optional), route ID, rating, comments, timestamp	Passenger mobile/web application

8. EXTERNAL INTERFACE REQUIREMENTS

8.1 User Interfaces

- Passenger Mobile/Web Application — real-time arrival, crowding levels, trip planning, and feedback submission.
- Transit Operator Dashboard — demand forecasts, schedule recommendations, and analytics visualizations.
- Fleet Manager Console — fleet allocation, vehicle status, and depot management views.
- Administrator Console — user management, system configuration, and system health monitoring.

8.2 Hardware Interfaces

- GPS receivers installed on buses/trains transmitting location data at regular intervals.
- IoT-based passenger occupancy sensors (e.g., infrared counters, weight sensors, camera-based counters).
- Depot servers / edge gateways aggregating vehicle telemetry before transmission to the cloud platform.

8.3 Software Interfaces

- Weather Data API (REST/JSON) — supplies current and forecast weather conditions.
- Traffic Data API (REST/JSON) — supplies real-time traffic and incident data.
- Existing Automatic Fare Collection (AFC) / ticketing system — supplies historical ridership counts.
- Notification Gateway (SMS/Push/Email) — delivers real-time alerts to passengers and operators.

8.4 Communication Interfaces

- HTTPS/TLS-secured REST APIs for all client-server and third-party integrations.
- MQTT or equivalent lightweight protocol for real-time IoT/GPS data streaming.
- Webhook-based event notifications for schedule changes and alerts.

9. SYSTEM FEATURES

The following key features summarize the system's primary capabilities and their relative implementation priority, based on their contribution to the core objectives of demand forecasting and service optimization.

Feature	Description	Priority
AI Demand Forecasting	Predicts passenger demand per route/stop/time-slot using historical and real-time contextual data.	High
Dynamic Schedule Optimization	Recommends optimized trip frequencies and timings based on forecasted demand.	High
Fleet Allocation Optimization	Assigns vehicles and drivers to routes to match predicted demand and minimize idle capacity.	High
Real-Time Passenger Flow Monitoring	Tracks live vehicle occupancy and location using IoT/GPS data.	High
Real-Time Passenger Information	Provides commuters with live arrival times and crowding levels.	High
Travel Pattern Analytics	Identifies recurring ridership trends to support long-term planning.	Medium
Transportation Analytics & Reporting	Generates configurable performance and utilization reports.	Medium
Real-Time Alerts & Notifications	Sends delay, overcrowding, and schedule-change alerts to users.	Medium
Feedback Management	Collects and analyzes passenger feedback on service quality.	Low
User & Role Management	Provides role-based account administration and access control.	Medium

9.1 High-Level System Architecture

The diagram below illustrates how data flows from external and on-vehicle sources through ingestion, AI/analytics processing, and application services, to the presentation layer used by each actor.

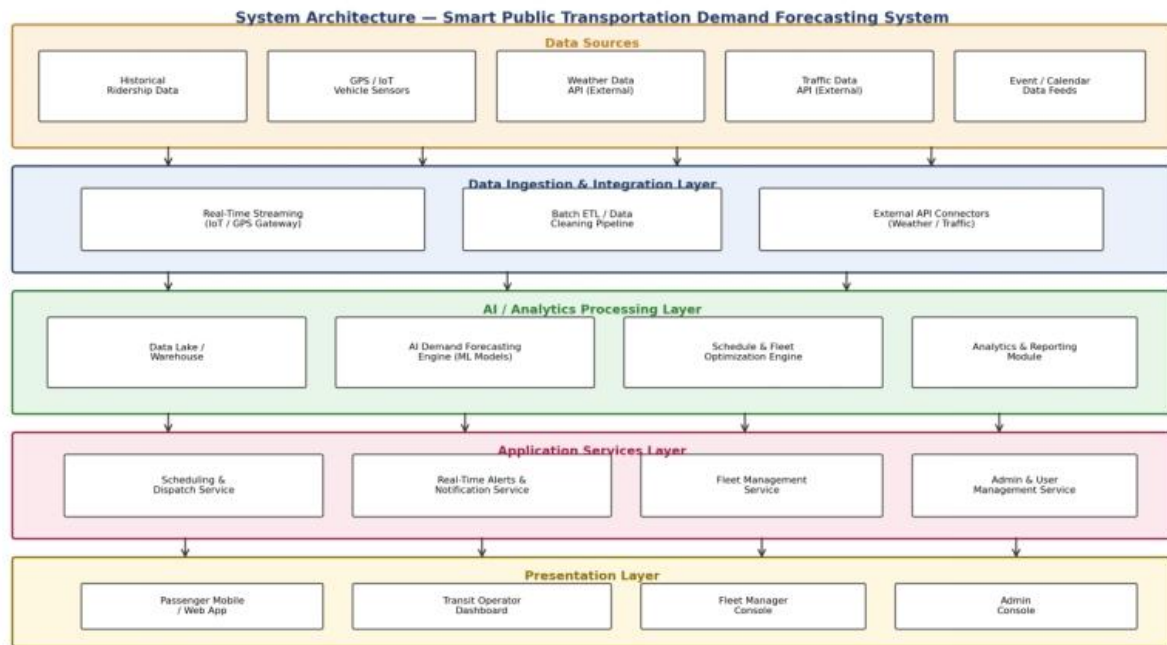


Figure 2: High-Level System Architecture

10. ASSUMPTIONS AND CONSTRAINTS

10.1 Assumptions

- Historical ridership data of sufficient volume and quality is available for initial AI model training.
- Buses and trains are equipped with, or can be retrofitted with, GPS and IoT occupancy sensors.
- Third-party weather and traffic APIs provide reliable, timely, and reasonably accurate data.
- Transit agency staff will be trained to interpret and act on AI-generated recommendations.
- Passengers have access to smartphones or web-enabled devices to use the real-time information app.
- Network connectivity (cellular/Wi-Fi) is available along the majority of transit routes for real-time data transmission.

10.2 Constraints

Type	Constraint
Technical	The system must integrate with the transit agency's existing legacy ticketing/AFC infrastructure, which may have limited API support.
Operational	Schedule and fleet changes recommended by the AI must still allow for manual override by human operators before going live.
Economic	The project budget limits the number of vehicles that can be retrofitted with new IoT sensors in the initial rollout phase.
Legal / Regulatory	The system must comply with regional data privacy regulations regarding the collection and storage of passenger location and usage data.
Time	The initial pilot deployment must be completed within one operating season to evaluate impact before peak-season scaling.
Environmental	Sensor and GPS hardware must operate reliably under varying outdoor weather conditions (heat, rain, cold).

11. REQUIREMENT VALIDATION — COMPLETENESS AND CONSISTENCY

11.1 Stakeholder Coverage Check

Each identified stakeholder group has at least one corresponding functional requirement and use case: Passengers (FR-05, FR-09, FR-12; UC-04), Transit Operators (FR-01, FR-02, FR-06; UC-01, UC-02), Fleet Managers (FR-03, FR-11; UC-03), Administrators (FR-10; system configuration use cases), and external data providers (FR-08). This confirms that no primary stakeholder's core need has been left unaddressed.

11.2 Ambiguity, Redundancy, and Inconsistency Review

- **Ambiguity:** Terms such as “real-time” and “optimized” were quantified with measurable targets in the Non-Functional Requirements (e.g., dashboard refresh ≤ 30 seconds, forecast MAPE $\leq 15\%$) to remove subjective interpretation.
- **Redundancy:** Schedule optimization (FR-02) and fleet allocation (FR-03) were kept as distinct requirements with an explicit `<<include>>` dependency, rather than merged, to avoid duplicated logic while preserving traceability between them.
- **Inconsistency:** Cross-checked that every functional requirement referencing forecasted data (FR-02, FR-03, FR-09) depends on FR-01, ensuring no requirement assumes data that another requirement does not produce.
- **Missing Requirements:** Added FR-13 (Model Retraining) and FR-14 (Audit Logging) after review, since an AI-driven system requires an explicit mechanism to keep predictions accurate over time and to maintain accountability for automated recommendations.

11.3 Suggested Improvements

1. Introduce measurable acceptance criteria (e.g., specific accuracy thresholds) for each AI-related requirement to simplify future testing and validation.
2. Add a formal data-quality validation requirement to reject or flag corrupted sensor/GPS readings before they reach the forecasting model.
3. Define a fallback/degraded-mode requirement for when external weather or traffic APIs are unavailable, beyond what is currently specified for forecasting alone.
4. Include explicit accessibility requirements (e.g., WCAG 2.1 AA compliance) for the passenger-facing application to ensure inclusivity.
5. Conduct a formal requirements walkthrough with transit operator and fleet manager representatives to validate that the proposed use cases match real operational workflows.

11.4 Traceability Summary

All eight stated objectives (forecast demand, optimize schedules, improve fleet allocation, monitor passenger flow, analyze travel patterns, generate analytics, reduce overcrowding, improve efficiency) are traceable to at least one functional requirement, one use case, and one

expected outcome, confirming that the requirement set is complete and internally consistent with the original problem statement.